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The Magnetism of Some Permian Rocks from New South Wales

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Summary

Rock samples of Permian age from Australia have a natural remanent magnetization consisting of components with a wide range of stability. The components of high stability are directed almost vertically downwards. Magnetite is the dominant magnetic phase, and this shows a range of properties which are related to the degree of stability observed. The steep inclination of the directions indicates a high latitude in Permian time for Australia, and this is consistent with the occurrence in the Permian System of beds of supposed glacial origin. The positive polarity is the same as that found in most Permian and uppermost Carboniferous beds situated elsewhere in the world, and this leads to the proposal of the name *Kiaman* for this period of time. The boundaries of the Kiaman Magnetic Interval do not coincide with geological time divisions so that already existing geological names are not applicable.

1. Introduction

This paper gives an account of field observations and laboratory analysis of the magnetic properties of igneous rock samples collected from the standard Permian sections in Australia. Previous work of a very preliminary nature has been reported by Mercanton (1926) and Irving & Green (1958). The purpose of this study is to attempt to estimate the direction of the Earth's magnetic field in Permian time in Australia since these igneous rocks may have acquired a thermoremanent magnetization in the direction of the field at the time of cooling. The question of whether or not these directions have been preserved through subsequent geological time depends on the magnetic stability of the rock and this has been

investigated by field and laboratory methods. The dominant magnetic components have been identified.

Table 1

The Permian System

Narrabeen Series	Lower Triassic
Upper Coal Measures	Permian
Upper Marine Series	
Lower Coal Measures	
Lower Marine Series	
Upper Kuttung	Upper Carboniferous

2. Geological levels studied

The Permian of New South Wales has 4 divisions (Table 1). Igneous rocks occur in the Lower and Upper Marine Series and the samples were collected from these. In the Lower Marine Series of the Hunter Valley (David 1907) samples were obtained from one site in a basalt flow. The bedding measured in adjacent sediments dips 22° at 70° true bearing.

The Upper Marine Series was sampled along the Illawarra coast where it is flat-lying or sub-horizontal (Harper 1915). The beds studied are thick latite flows and associated intrusives interbedded with waterlaid tuffaceous sediments. The sampling sites are spread over about 50 km south of Wollongong. The sequence at any one place does not exceed about 450 m in thickness. The thickness of individual members varies laterally. The lowest member is the *Blowhole Flow*, which is 45 m thick near Kiama, from which samples were obtained from 5 sites over a distance of 8 km. The *Bombo Flow*, which overlies the Kiama Tuff, which in turn overlies the Blowhole Flow, has a maximum thickness of 150 m. Samples were obtained from 6 sites spread over a distance of 11 km. The *Dapto Latite* which is typically coarse-grained has a maximum thickness of over 75 m. It intrudes tuffaceous sediments which overlie the Bombo Flow. It was sampled at 3 sites over a distance of about 1 km near Port Kembla. The *Cambewarra Flow*, which has a maximum thickness of 180 m, was sampled at 3 sites over a distance of 10 km. The *Berkeley Latite* was sampled at 1 site through much of its maximum thickness of 25 m.

The sites studied are listed in stratigraphic sequence (oldest first) in Table 2. Radioisotope age determinations (K/A methods using plagioclase felspars) on the Bombo Flow (site L7) gave 244 My, and on the younger Berkeley Latite (site *R*13) gave 252 My (Evernden & Richards 1962, ref. Nos. GA 138 and GA 136). The age difference of 8 My is less than the error. These estimates suggest a Middle Permian age on Kulp's 1961 time scale (Kulp 1961) consistent with the geological evidence.

Table 2

Reference List of Collecting Sites

The reference number or letter is given in the left hand column. Sites at which oriented samples were collected and studies of the direction of natural remanent magnetization made are numbered. Sites at which the ferromagnetic components were studied are labelled by letters (letters I and O are not used). At many sites both types of study were made and these are labelled by a number and a letter. The name of the rock number is given together with the geographical coordinates and the grid-references on the 1 mile to the inch Military Map Series of Australia, the map sheet name being given in brackets.

LOWER MARINE SERIES (1 site)

A1 Basalt Flow, Comerford's Quarry; 32°41′ S, 151°34′ E; 587 588 (Paterson).

UPPER MARINE SERIES (18 sites)

B Blowhole Flow, Princes H'way road cutting; 34°42′ S, 150°50′ E; 844 161 (Kiama).

C2 Blowhole Flow, Princes H'way road cutting; 34°44′ S, 150°49′ E; 821 101 (Kiama).

3 Blowhole Flow, railway cutting; 34°44′ S, 150°49′; 825 106 (Kiama).

D4 Blowhole Flow, Princes H'way road cutting; 34°43′ S, 150°50′ E; 835 132 (Kiama).

E5 Blowhole Flow, Blowhole Pen. Kiama; 34°40′ S, 150°52′ E; 867 184 (Kiama).

F Bombo Flow, Princes H'way road cutting; 34°35′ S, 150°50′ E; 830 298 (Kiama).

G Bombo Flow, Shellharbour quarry; 34°36′ S, 150°53′ E; 881 292 (Kiama).

H Bombo Flow, Dunmore quarry; 34°37′ S, 150°50′ E; 828 258 (Kiama).

J Bombo Flow, Minnamurra quarry; 34°38′ S, 150°52′ E; 158 235 (Kiama).

K6 Bombo Flow, Kiama quarry; 34°40′ S, 150°51′ E; 847 188 (Kiama).

L7 Bombo Flow, Bombo quarry; 34°39′ S, 150°51′ E; 852 208 (Kiama).

M8 Dapto Latite, Port Kembla quarry; 34°29′ S, 150°54′ E; 893 416 (Wollongong).

N Dapto Latite, building excavation; 34°29′ S, 150°53′ E; 892 418 (Wollongong).

P9 Dapto Latite, road cutting; 34°29′ S, 150°53′ E; 886 415 (Wollongong).

10 Cambewarra Flow, road cutting; 34°47′ S, 150°34′ E; 566 038 (Nowra).

11 Cambewarra Flow, road cutting; 34°45′ S, 150°36′ E; 603 081 (Nowra).

Q12 Cambewarra Flow, road cutting; 34°46′ S, 150°39′ E; 654 068 (Nowra).

R13 Berkeley Latite, American Creek quarry; 34°46′ S, 150°51′ E; 840 464 (Wollongong).

3. Directions of natural remanent magnetization (M_n)

Several (n) cylindrical specimens were cut from each sample. The specimens are 3·5 cm in diameter and either 3·5 cm or 0·4 cm in thickness. The directions (often referred to for brevity as the initial directions) and intensities of the natural remanent magnetization of all specimens were measured on astatic magnetometers using the methods described previously (Collinson & others 1957). Specimen directions were measured to an accuracy of 3° and intensities to 5 per cent. The initial directions observed in specimens from the same sample, or in samples from the same site often do not agree very well, divergencies of 60° being common. Within a site, these scattered directions often have an approximately great circle distribution whose long axis, when continued, passes through the present or dipole field (Figure 1(a) and (c)); there is always a considerable scatter about this great circle. In other cases the directions have a high and irregular scatter (Figure 1(e)). Such distributions could be due to components of viscous magnetization imposed by the present Earth's field when the rock was *in situ* or during storage in the laboratory. Partial demagnetization was carried out to see if such low stability components were present (Thellier & Rimbert 1954, 1955).

4. Demagnetization by alternating magnetic fields

The apparatus and methods used have been described previously (Irving, Stott & Ward 1961). Peak field values ($H_p \sim$) are quoted throughout. The effects of demagnetization of single specimens are considered first. Changes in intensity observed indicate a wide range of stability (Figure 2). In some cases (sites A1, Q12) a rise of intensity indicates the removal of a component of low stability directed at more than 90° from that of a more stable component. An example of the way in which the magnetization vector changes is given in Figure 3. The

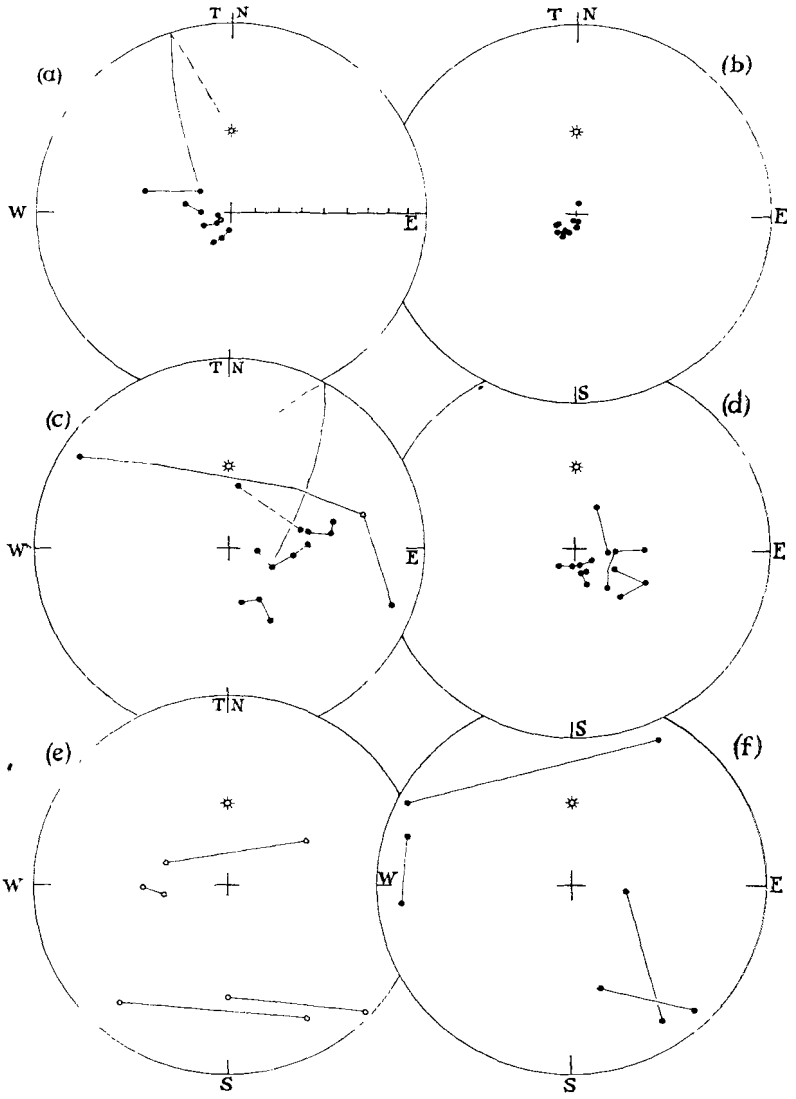


FIG. 1.—The directions of magnetization at 3 sites. Polar projections are used, the directions of the geocentric axial dipole field being indicated by a star. The primitive is the horizontal at the site. In all cases adjacent sediments are horizontal. Directions of specimens from the same sample are linked together except in (b) where this would lead to confusion. This figure illustrates the effect of partial demagnetization by alternating magnetic fields ($H_p \sim$), (a) Blowhole Flow site D4, initial directions, (b) same after partial demagnetization in 150 $H_p \sim$, (c) Blowhole Flow site C2 initial directions, (d) same after partial demagnetization in 150 $H_p \sim$, (e) Bombo Flow site L7 initial direction, (f) same after partial demagnetization in 300 $H_p \sim$.

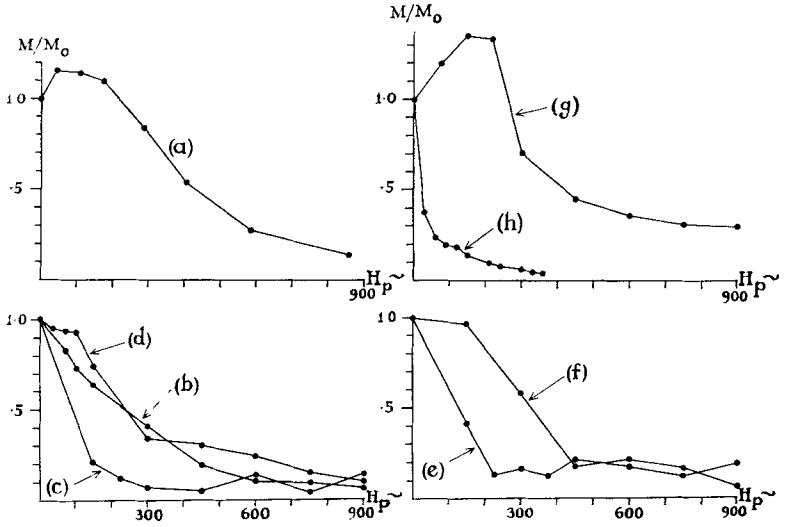


FIG. 2.—Alternating field demagnetization curves. The ratio (M/M_0) of the intensity of magnetization after treatment in a given field is plotted against that field. (a) Lower Marine basalt, site A1; (b), (c), (d) Blowhole Flow, site C2 (b), site E5 (c), site D4 (d); (e) Bombo Flow, site L7, (f) Dapto latite, site M8; (g) Cambewarra Flow, site Q12, (h) Berkeley latite, site R13.

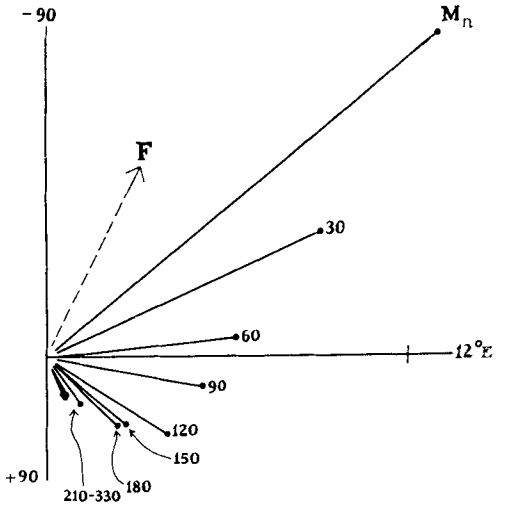


FIG. 3.—Rotation of magnetization vector caused by partial demagnetization. The projection of the vectors on the plane of the magnetic meridian are plotted. They are labelled with the appropriate alternating field strengths. The length marked out on the abscissa is equivalent to an intensity of 10×10^{-5} emu/cm^3 or $3{\cdot}7 \times 10^{-5}$ emu/g.

initial direction (13, −41) is 24° away from the present field direction **F** (12, −65) Progressive treatment in steps of 30 Oe caused a rotation of about 100° along the path of a great circle through the initial direction and **F** and progressively away from **F** until a direction (12, +60) was reached in fields of 210 Oe. No systematic changes occurred in higher fields. Two components are present, one of low stability ($\mathbf{M}_u$) directed along **F**, and one of high stability ($\mathbf{M}_{st}$) with a steep positive inclination. Since the direction and magnitude of $\mathbf{M}_n$ and the directions of $\mathbf{M}_{st}$ and **F** are measured the magnitudes M_{st} and M_u can be calculated (Table 3).

Table 3

Magnetic properties

Values for individual samples are given because of variability within sites. χ is the initial susceptibility, the ratio of the induced magnetization in 0·49 Oe to that field; M_n is the intensity of the natural remanent magnetization; M_{st}, M_u are the magnitudes of the stable and unstable components of M_n; σ_s is the saturation magnetization, θ is the Curie temperature (the values in brackets being those obtained after heating). Magnitudes are given in emu per gram × 10^3.

	Site number	Sample number	χ	M_n	M_{st}	M_u	σ_s	Principal component	Equivalent Fe_3O_4	Minor components		Unit cell
								θ° C	per cent	θ° C	Fe_3O_4 per cent	Å
Berkeley Latite	R13	117	—	0·05	—	—	—	—	—	—	—	—
		119	—	0·05	0·06	0·09	1400 (1300)	550 (540)	1·5	Variable	<0·2	—
Cambewarra Flow	Q12	108	—	0·06	0·17	0·10	1100 (1600)	570 (565)	1·2	350	<0·1	8·39
	11	103	—	0·05	—	—	—	—	—	—	—	—
	10	113	—	0·13	—	—	—	—	—	—	—	—
Dapto Latite	P9	7	0·71	0·09	—	—	2400 (2200)	530 (510)	2·7	500 (420)	0·6	—
		8	0·35	0·30	0·30	negligible	1100 (1100)	560 (560)	1·2	500,400,350	0·1	8·38
		47	—	0·16	—	—	—	—	—	—	—	—
	N	5	0·45	0·29	—	—	1500 (1600)	560 (530)	1·7	300	0·2	8·39
		6	0·85	0·10	—	—	2200 (1900)	500 (450)	2·4	—	—	—
	M8	9	0·38	0·33	0·33	negligible	1600 (1800)	560 (560)	1·8	absent	—	8·39
Bombo Flow	L7	18	1·21	0·16	—	—	2400 (1800)	480 (420)	2·7	—	—	—
		125	—	0·12	—	—	—	—	—	—	—	—
	K6	16	—	—	—	—	3000 (2600)	500 (450)	3·3	400	<0·2	—
		120	—	0·18	0·22	0·33	—	—	—	—	—	—
	J	20	1·10	0·14	—	—	2800 (2100)	500 (460)	3·1	absent	—	—
	H	21	1·10	0·14	—	—	2600 (2100)	490 (430)	2·9	400	<0·2	8·39
	G	23	1·10	0·39	—	—	3100 (2800)	540 (540, 490)	3·4	350	<0·2	—
		24	1·20	0·38	—	—	3600 (3200)	540 (530)	4·0	500, 350	<0·5	—
	F	25	1·10	0·04	—	—	2600 (1800)	500 (380)	2·9	550 (510)	<0·5	—

Table 3—*continued*

Blowhole Flow	E	5	1	1·10	0·25	0·98	0·80	3000 (3300)	555 (545)	3·3	absent	—	8·39
			2	1·70	0·24	1·40	1·38	3600 (3300)	530 (500)	3·6	350	<0·4	—
			30	—	0·14	0·30	0·39	3700 (3000)	510 (450)	4·0	absent	—	—
			31	—	0·23	0·30	0·11	3400 (2800)	510 (440)	3·7	variable	<0·4	—
			32	—	0·26	—	—	3500 (2800)	510 (440)	3·8	absent	—	—
			33	—	0·44	—	—	—	—	—	—	—	—
	D	4	104	—	0·75	0·87	0·30	4000 (3600)	560 (560)	4·0	absent	—	—
			45	—	0·47	—	—	—	—	—	—	—	—
			105	—	0·41	—	—	—	—	—	—	—	—
	C	2	37	—	0·22	—	—	—	—	—	—	—	—
			41	—	0·25	—	—	—	—	—	—	—	—
	B		4	—	0·13	0·27	0·17	2600	565 (565)	3·0	absent	—	—
			13	1·41	0·14	—	—	3500 (2800)	490 (460)	3·9	450	<0·4	8·40
			14	1·02	0·13	—	—	3700 (3000)	500 (450)	4·1	variable	<0·2	—
Lower Marine Basalt	A1		26	0·75	2·30	2·80	0·43	1100 (1400)	565 (565)	1·2	absent	—	8·39
			205	—	0·41	—	—	—	—	—	—	—	—

Table 4

Direction of magnetization

S is the number of samples, N the number of specimens cut from these. D, I are the declination and inclination of the site means. R_S is the resultant of sample means, R_N of specimen directions. R_n is the resultant from specimens for each sample. $H_p\sim$ is the strength of the alternating magnetic field used.

	Collecting site	S	N	Natural remanent magnetization					After treatment			
				D, I	R_S	R_N	R_n	H_p	D, I	R_S	R_N	R_n
R13	Berkeley Latite		13	random	1·19	4·56	11·73	225	336, +86	3·13	10·19	12·16
Q12	Cambewarra Flow	5	13	242, +54	4·89	12·63	12·89	225	211, +72	4·95	12·74	12·84
11	Cambewarra Flow	2	9	214, +16	1·99	8·69	8·75	225	217, +72	1·99	8·82	8·85
10	Cambewarra Flow	2	11	293, −27	1·69	8·36	9·85	225	255, +48	1·89	10·05	10·64
P9	Dapto Latite	4	11	108, +14	2·65	5·37	8·38	150	136, +77	3·14	8·71	10·55
M8	Dapto Latite	3	15	276, +75	2·68	13·41	13·88	150	225, +76	2·82	14·03	14·39
L7	Bombo Flow	4	8	random	3·01	5·39	7·16	300	random	1·63	2·95	6·95
K6	Bombo Flow	2	12	325, −53	1·68	9·55	11·39	300	225, +70	1·77	6·49	7·23
E5	Blowhole Flow	5	15	344, +37	2·27	4·44	12·59	150	30, +86	4·52	13·12	14·59
D4	Blowhole Flow	4	11	266, +77	3·93	10·77	10·94	150	221, +83	3·98	10·95	10·99
3	Blowhole Flow	3	13	327, +15	1·70	7·62	12·28	150	325, +86	2·96	12·66	12·82
C2	Blowhole Flow	5	15	82, +54	4·15	11·55	12·86	150	117, +77	4·89	14·59	14·88
A1	Lower Marine Basalt Flow	6	12	248, +77	5·91	11·60	11·77	150	230, +76	5·97	11·88	11·93

In some cases M_u is negligible and $\mathbf{M}_n$ is stable (P7, M8). In others (E5(2)) the components $\mathbf{M}_u$ and $\mathbf{M}_{st}$ have similar magnitudes but being almost antiparallel yield a small natural remanence.

The effects of partial demagnetization on the direction of magnetization of groups of specimens from different sites are illustrated in Figure 1. The results in (b) and (d) show the way in which great circle distributions or scattered directions condense about a direction of high magnetic stability. The directions at site L7 (Figure 1(e)(f)) are random and no stable magnetization is detectable; partial demagnetization redistributes the directions without reducing their scatter, and it seems possible that the decay of natural remanence with time has approached completion, the minimum "noise" level due to statistical fluctuations only remaining. The irregular form of the demagnetization curve (Figure 2(e)) in high fields supports this view, the rapid initial fall being due to the removal of a temporary component acquired during storage in the laboratory (this may be reacquired by repeating the storage procedure for a period of a year). At the other sites there is an underlying component of high stability which is partly masked by low stability components. It is desirable to remove the latter and this has been achieved by partial demagnetization of all specimens by alternating magnetic fields.

The procedure is to treat 4 pilot specimens (where possible from different samples) from a site in an alternating field ($H_p \sim$) of 150 Oe in the absence of a steady field and determine the direction and intensity of magnetization. The treatment is repeated, usually in steps at 225, 300 Oe etc. until minimum dispersion of the directions of the 4 specimens is achieved. All specimens at the site were then treated in that field. The required field, which was always between 150 and 300 Oe, was determined separately at each collecting site. Treatment in fields greater than this caused the directions to scatter somewhat but their mean directions were not altered significantly.

The mean site directions before and after partial demagnetization are listed in Table 4 and plotted in Figure 4. At all sites except L7, at which the directions are random, partial demagnetization improves the agreement between observations at a site. This is seen by comparing the lengths of the vector resultants, R. The mean site directions initially are randomly distributed. After partial demagnetization they are grouped closely together with a steep positive inclination. It is uncertain whether or not the differences between mean site directions (Figure 4(b)) are significant. This is a question of general importance since the estimates of the between-site dispersion can, in principle, be used to study the variation of the magnitude of the secular variation with latitude, and thus of testing the hypothesis that the Earth's field has always been dipolar. This is a fundamental problem in palaeomagnetic work. But the analysis of dispersion of observations of unequal value (that is, variable within-site scatter such as is observed here) has not yet been devised. For this reason the basic statistical data are entered in Table 4 to facilitate future analysis; these comprise the vector resultants obtained by combining observations at different levels.

The mean direction (232, +81) of the stable magnetization of the Upper Marine Latites has been obtained as a mean of the site directions giving each unit weight. Fisher's (1953) analysis is used. The circle of confidence is 6° ($P = 0{\cdot}05$). This result is based on 138 specimen observations, 43 oriented samples ($R_{\text{samples}} = 35{\cdot}01$) and 11 sites ($R_{\text{sites}} = 10{\cdot}62$).

The Triassic sediments (Table 1) overlying the Permian on the Illawarra coast have directions with steep negative inclination approximately reversed from those

in the Permian. The reversal occurs between sites R13 and the Chocolate Shales of the Narrabeen Series (site CS4, Irving 1963a). These sites are approximately 100 m apart stratigraphically, the sequence between them being conformable. For convenience of description we propose to call this reversal the *Illawarra Reversal*.

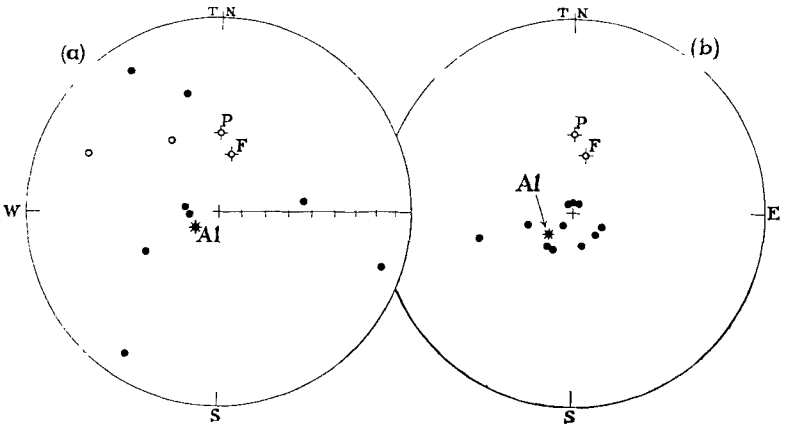


FIG. 4.—Mean site directions. Polar projections are used, the primitive being the bedding at each collecting site. Dots indicate positive inclinations plotted on the lower hemisphere and circles are negative inclinations plotted on the upper hemisphere. The direction labelled A1 is from the Lower Marine basalt. The remaining directions are from the latites of Upper Marine Series. (a) Initial directions. Directions from sites L7 and R13 (random) are not plotted, (b) Directions after partial demagnetization. Site L7 (random) has no significant mean direction.

5. Thermal demagnetization

Thermal demagnetization was also used to study stability at one site (Q12). The initial directions of 4 pilot specimens, one from each of 4 oriented samples, are plotted in Figure 5(a), and the directions after heating to 300 °C and cooling in zero field are plotted in Figure 5(b). For comparison results from 4 other specimens from the same 4 oriented samples initially and after partial demagnetization in an alternating magnetic field of 225 Oe are plotted in Figure 5(c) and (d). Both treatments produce similar effects—the precision improves and the magnetization vectors rotate away from present field direction. The mean direction for each pilot set of 4 specimens is 223, +66 for thermal demagnetization, and 227, +72 for demagnetization in alternating fields; these are not significantly different. Progressive demagnetization at higher temperatures and fields did not change these mean directions significantly.

6. Magnetic minerals: thermomagnetic analysis

Thermomagnetic analysis was carried out for the most part on whole rock specimens because of the difficulty of preparing representative mineral extracts. In thin section the iron minerals are found to occur in a wide range of sizes from tiny particles in the fine grained ground mass to crystals up to 0·5 mm across. When extracts were obtained with a hand magnet (after crushing and sizing) heavy contamination by other minerals always occurred. Further preparation would lead to greater selectivity and for this reason no attempt was made at chemical analysis.

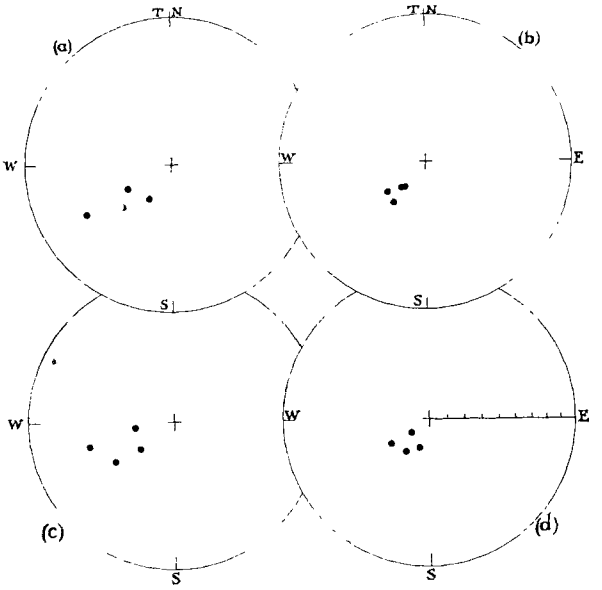


FIG. 5.—Thermal and alternating field demagnetization compared. (a) and (c) are the directions of natural remanent magnetization in 8 specimens from site Q12. (b) and (d) are the directions after partial demagnetization at 300 °C (b) and 225 Oe (d).

Thermomagnetic analysis of whole rock fragments (approximately 1 g) was carried out for one or more samples at the sites listed in Table 3. The measurements were made with a pendulum magnetometer (Parry 1960) where the saturation magnetic moment (σ_S) induced by a magnetizing field of approximately 5000 Oe is measured as the temperature is increased to 650 °C and then decreased. The value of σ at temperature T°C (σ) divided by σ at 20 °C (σ_S) is plotted against the temperature, and typical values are plotted in Figure 6.

Usually the specimen was heated in an evacuated silica tube sealed at an initial pressure of less than 0·1 mm Hg. As the specimen temperature was increased some outgassing occurred and the partial pressure of the residual gases was unknown at high temperature. Results were comparable for specimens prepared in this fashion, for specimens of the same rock heated in air, and for specimens of magnetic extracts from the same rock sealed under higher vacuum conditions. When specimens were heated in air however some magnetic material was converted to a weakly magnetic form which was shown by X-ray analysis in one case to be hematite. For fine textured rocks the value of the induced saturation magnetic moment was reproducible from specimen to specimen from the same sample to within 10 per cent. In the coarse grained samples the specimens were too small to be a representative sampling so values between specimens from the same sample vary by a factor of 2 or 3 and in these cases an average of 3 or more specimens per sample has been given. Curie points are taken as the points of inflexion of the thermomagnetic curves and such points are reproducible to within 5 °C for standard specimens of iron or nickel. In the rock specimens the Curie point can be established to $\pm$ 5 °C for a major component and to $\pm$ 10 °C for most minor components.

Details for particular samples are listed in Table 3. The magnetic minerals may be divided into 2 major types.

Type 1. This material has a Curie point of (560 ± 10) °C which is reproducible after heating to 650 °C. The saturation magnetic moment at room temperature is usually increased after this heating cycle (see Figure 6(a) Curves 1, 2). Progressive reduction in the Curie point is noted when a specimen is submitted to temperatures greater than 900 °C.

Type 2. This material has a Curie point of (490 ± 20) °C which is reduced by more than 40° after heating to 650 °C (Figure 6(a) Curves 3, 4). The amount by which the Curie point is diminished depends on the heating cycle (Figure 6(b)). Little change in the saturation moment (room temperatures) is noted if the heating has not exceeded 400 °C but after heating to between 400 ° and 600 °C this is normally reduced.

There is no obvious relationship between the texture of the rocks and the occurrence of a particular type of magnetic constituent. Any particular specimen usually has a major magnetic component belonging to one or other type. Single components only occur in samples from sites E5, A1, C2, J and M8. In estimating the amount of magnetic mineral in a sample it was assumed that the saturation moment of the pure magnetic mineral was 92 emu/g. The estimated amounts may be too low (Table 3) but they are included since they give the relative proportions correctly. Minor components with lower Curie points are observed at other sites. These minor components are variable from specimen to specimen in a sample and are changed by heating, but rarely comprise more than 10 per cent of the total ferromagnetic content.

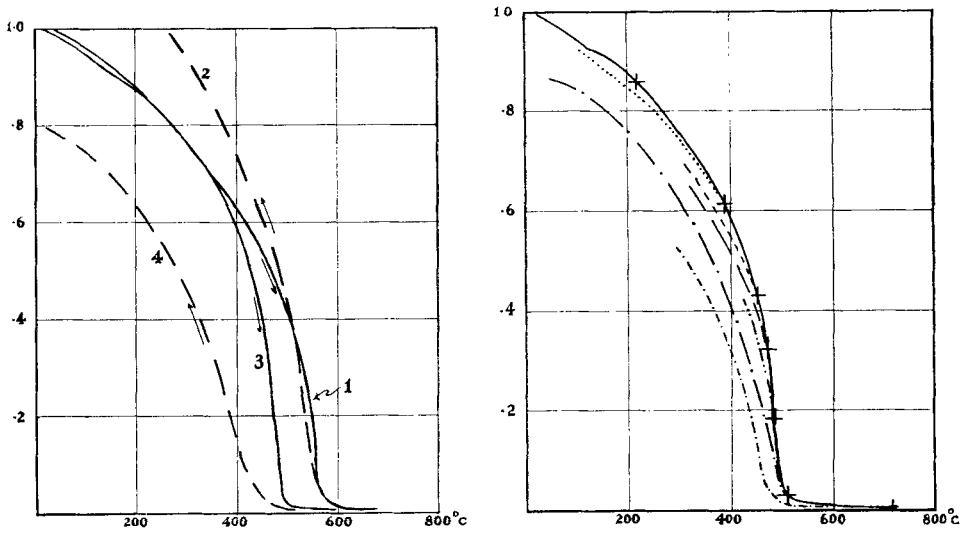


FIG. 6.—Thermomagnetic curves. (a) Curves 1, 2 are heating and cooling curves for sample number 9 (Type 1 "magnetite"). Curves 3, 4 are for sample number 21 (Type 2 "magnetite"). (b) A series of curves obtained by heating the sample to a temperature less than the Curie point and then cooling, this cycle being repeated for increasing maximum temperatures as indicated by + (Type 2 "magnetite", sample 21).

7. Magnetic minerals: X-ray analysis

Magnetic extracts were taken from samples from 7 sites. These extracts were examined by the Debye-Scherrer technique and the cell size determined. The same extracts were analysed thermomagnetically and there was some evidence that

the separation had been selective, but the general characteristics of the whole rock were present also in the extracts. Mineral extracts gave complex powder patterns of lines but the prominent lines were consistent with the face centred cubic structure of magnetite. The unit cell size $8{\cdot}39 \pm 0{\cdot}01$ Å, is larger than usually found for pure magnetite (ASTM Crystallographic index). In each case the lines were rather broad thus limiting the precision of the measurements and indicating some crystalline imperfection or non-uniform composition. Within the measurement errors no significant difference in unit cell dimension was found between types 1 and 2 or between specimens of either type before and after heating to 650 °C. Lines which did not belong to magnetite were consistent with the major lines of the non-magnetic minerals in the rock, such as pyroxene, and were due to impurities in the extract. On the basis of the thermomagnetic and X-ray analysis the dominant ferromagnetic mineral is classified as "magnetite".

8. Other magnetic properties

The initial susceptibility (χ) was determined using an astatic magnetometer (Collinson & others 1957) for 16 specimens. The values of are the same order as the natural remanence (M_n) in those cases where the magnitude of the unstable components are low compared with that of the stable component and in which minerals of Type 1 occur (Table 3). In other cases the ratio (χ/M_n) falls as low as one-tenth. This is to be expected on general grounds since χ depends on the amount of magnetic material present, and M_n on the degree of internal alignment which may be expected to be greater in the more stable specimens. Seventeen determinations of the coercivity of maximum remanence (Creer & others 1959) gave an average of 260 and a range of 210–380 Oe.

Stacey (1960) has shown that the bulk magnetic anisotropy (anisotropy of saturation magnetization) of 2 specimens from the Upper Marine latites is small. Five representative specimens from the collections were heated to 570 °C and cooled in the Earth's field and the magnetization acquired was parallel to that field. Specimens of these Permian igneous rocks heated to 650 °C and cooled in a field and an axial load of 500 kg/cm^2 acquired a magnetization parallel to the field (Stott & Stacey 1959).

9. Discussion

Four points of interest arise from these results.

(1) Specimens with a high Curie point are more stably magnetized than those with a lower Curie point (less than 540 °C). The rocks least affected by unstable components are those which contain reversible Type 1 "magnetite". It is not obvious that this should be so since stability in alternating fields depends on the coercivity which is related to grain-size, whereas the Curie temperatures depend on chemical composition. For removal of unstable components, and as an alternative to alternating field demagnetization, thermal demagnetization can be used safely on high Curie point rocks providing the temperature maximum is less than 400 °C. In rocks with a low Curie point some change does occur in σ_s after a specimen is heated beyond 300 °C (Figure 6(b)) but there is no appreciable change in Curie point unless this temperature exceeds 600 °C, and it appears that below this temperature the mineral is chemically stable.

(2) After partial demagnetization a direction of considerable uniformity and stability remains. The direction in the Lower Marine basalt, after correction for

tilt, agrees closely with those in the sub-horizontal Upper Marine latites. The tilting is pre-Triassic. A Permian age for the stable magnetization is indicated. The thermal stability of the magnetic minerals in certain cases suggests that any heating during burial will have had little effect. The absence of any substantial bulk anisotropy, and the fact that thermoremanence induced in the laboratory either with or without stress is parallel to the applied field, indicates that the magnetization acquired when the rocks cooled in the first place was parallel to the ambient field. For these reasons, and because of the uniformity observed through a thick rock sequence, it is reasonable to suppose that the stable directions (Figure 4(b)) are those of the Earth's magnetic field at the time the rocks were formed, that is, in Permian time.

(3) The results are of some palaeoclimatic interest. The Upper Carboniferous and Permian beds in Australia contain evidence of extensive glaciation and the question arises, was this due to Australia being near the geographic pole, or was it due to the occurrence of cold climates in comparatively low latitudes at that time? Steep inclination (84°) which suggests a high latitude and which supports the former

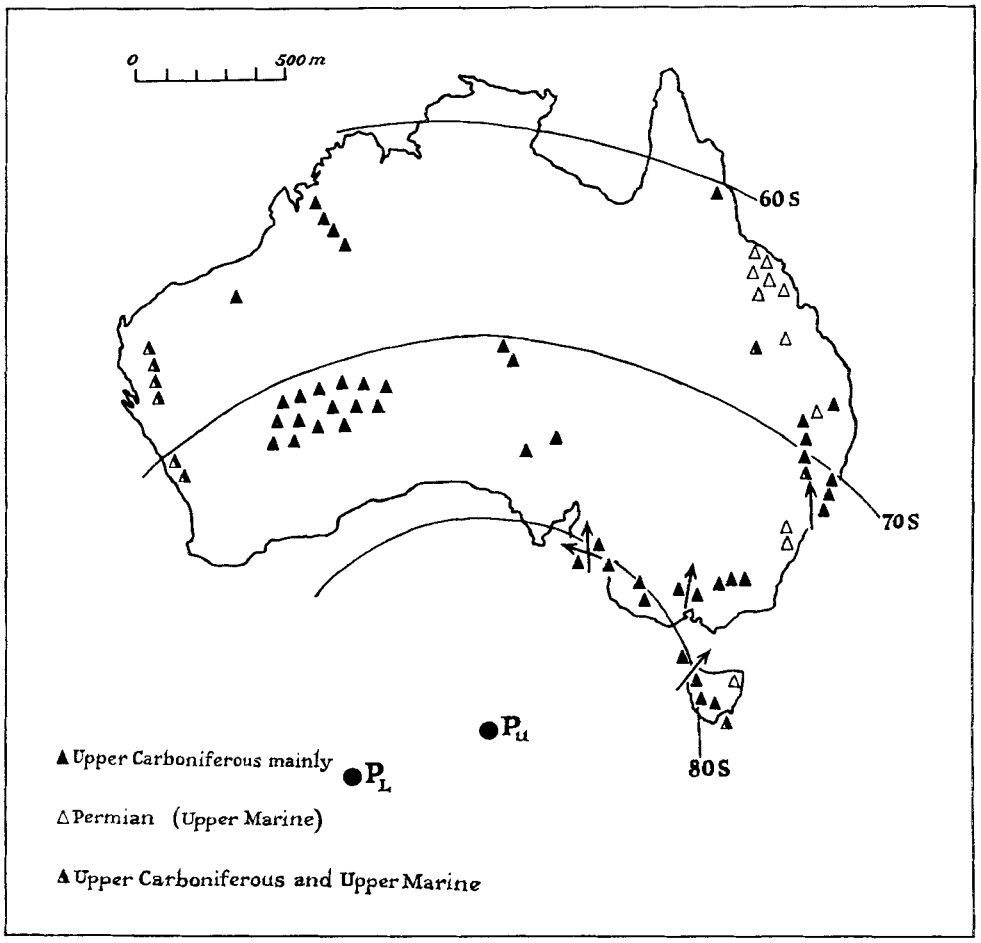


FIG. 7.—Latitude of Australia in the Permian. P_L and P_u are poles obtained from the Lower and Upper Marine results. The direction of ice movement is indicated by arrows and occurrences of evidence of glaciation by triangles taken from David (1950, Figure 116).

alternative have already been recorded from Upper Carboniferous glacial beds (Irving 1957). The present work extends this record into the overlying Permian beds so that the palaeomagnetic results now span, in a stratigraphic sense, the entire glacial period in Australia. David and Sussmilch (1931) recognized glacial horizons in both the Lower and Upper Marine Series of N.S.W., and it is from these *same* sequences that the samples for magnetic work have been collected. It is therefore perhaps not fortuitous that the observed inclination of the stable component is high.

In order to effect a comparison with glacial evidence elsewhere in the country the palaeomagnetic pole is calculated from the mean directions assuming a dipole configuration for the Earth's field: Upper Marine latites, mean location 34°38′ S, 150 47 E, pole 44 S, 132 E; Lower Marine latites, location 32°41′ S, 151 34 E, pole 46 S, 122 E. The first is based on sampling of a thick igneous sequence with interbedded sediments and may therefore be expected to have averaged out the secular variation. The second is based on a single flow. The two are in close agreement and do not differ significantly. They are plotted in Figure 7 together with the evidences of glaciation and supposed direction of ice movement. The parallels of latitude relative to the Upper Marine pole (being the more firmly based) are plotted. The glacial deposits lie mostly between 60 and 80° S (palaeolatitude). This result supports the idea that the Earth's field has had an axial dipole form in the Permian as in Recent times, and is also consistent with the view that the Permian glaciation arose from Australia being in a high southerly latitude. The palaeomagnetic poles are about 90° of arc from those obtained from Europe and North America.

(4) The stable directions of magnetization all have steep positive inclinations. Many authors would refer to these as *reversed* magnetizations by analogy with the reversed magnetizations commonly encountered in Pleistocene and Upper Tertiary beds where the directions are approximately opposed to the present dipole field. But this is confusing in the present context since the axis of magnetization is 30° from the present dipole field. The convention used here is to regard the directions as negative (positive) when the pole towards which the north (south) seeking magnetization points is in the present day northern hemisphere. The convention is adequate for comparing polarities observed in Carboniferous, Permian and older Triassic beds from different parts of the world and this is done in Table 5. All beds of uppermost Carboniferous age so far studied have positive polarization, their directions being undistinguishable (within the errors) from those observed in overlying Permian beds. The mean directions of all beds studied of Permian age, except the Upper Tartarian of Russian (Khramov 1961) are positive. The Upper Tartarian, which is sometimes classified with the Triassic by geologists, has both positive and negative or mixed polarities. Negative or mixed polarities characterize Lower Triassic beds elsewhere in the world. Beds laid down in the time range uppermost Carboniferous to uppermost Permian are characterized by having mean directions with positive polarity. This is not statistical; all the formations studied show this feature and Table 5 shows that these give a fair coverage of this span of time, so that although further sampling may reveal negative directions the chances of them being of more than subsidiary importance is considered unlikely. This span of time does not coincide with the geological Permian period and there is, of course, no reason why it should do so. In western Europe the change from negative to positive appears to occur within the Westphalian or between the Westphalian and Stephenian although the actual level has not yet been identified on the ground. A level of reversal which may correspond

Table 5

Magnetic polarities of Carboniferous, Permian and older Triassic rocks

This table gives the sequence of polarities observed in the U.S.S.R., Western Europe, North America and Australia. The rock ages are denoted by C (l, m, u) Carboniferous (lower middle and upper), P (l, m, u) Permian, and Tr (l, m) Triassic. Results from Upper Triassic beds are not included. In some cases, as for example, in the Australian section and in the Upper Permian of Russia, the beds are arranged in stratigraphic sequence as shown by superimposition evidence. Elsewhere the relative ages of beds within the same region (for instance, many from the Permian of Europe) is less clear and the order given here is arbitrary. The detailed correlations between regions is subject to much uncertainty and is made here on the basis of the ages assigned in originals. The reference numbers refer to the bibliography compiled in this journal (**3**, 96 and 444; **5**, 70; **6**, 263; **7**, 263).

	U.S.S.R.			Western Europe			North America			Australia			
TRIASSIC	Tr	Siberian traps (5/27–29; 3/31, 32)	M										
	Tr	Intrusives Siberia (5/30–32)	M				Trl–m	Moenkopi Fm. (3/37–44)	M				
	Tr	Sediments Taiymur pen (5/33)	—	Trl	Bunter (4/8–12)	M							
	Trl	Sediments (2/32; 5/23–26)	M	Trl	Vosge ss. (1/63; 4/7)	M	P–Tr	Chugwater Fm. (4/45–54)	M	Trl	Narrabeen sediments	—	
PERMIAN	Pu	Up. Tartarian (5/38–40)	M								*Illawarra Reversal* 100*m*		KIAMAN
	Pu	Lr. Tartarian (5/41–43)	+										
	Pu	Kazanian (2/34; 5/44)	+										
	Pu	Ufimian (2/35; 5/45)	+										
	Pm–u	Maymecha-Kotuy Ultrabasics (3/55)	+	Pm	Moutcenis Ss. (1/83)	+				P(m–u)	Upper Marine latites	+	
				P(m?)	Nideck porphyry (1/82)	+	P(l–m)	Yeso Fm. (1/85)	+				
				P	Nahe Igneous rocks (2/37; 4/36)	+							
				P	Mauchline beds (1/74; 75)	+							
				P	Ayrshire kylites (1/76)	+							
				P	Esterel suite (1/78–81)	+	P	Culter Fm. (1/86)	+				
				P	Exeter traps (1/73)	+							
				Pl	Oslo igneous complex (5/37)	+	Pl	Abo Fm. (1/87)	+	Pl	Lower Marine latites	+	
	Pl	Donbas sediments (5/46)	+	Pl	St. Wendel Ss. (1/84)	+							
				Pl	Rotliegende (4/15–18)	+	Cu–Pl	Supai Fm. (1/88–91; 3/56–58)	+				
				Cu–Pl	Whin Sill (2/36)	+	Cu–Pl	Sangre de Cristo Fm.	+				
CARBONIFEROUS	Cu	Arancarite stage (5/48)	+	Cu	Stephanian (4/23–25)	+	Cu	Naco Fm. (1/105; 3/59)	+				
				Cu	Midland sills (4/20)	+				Cu	Upper Kuttung glacials	+	
	Cu	Avilov stage (5/49)	+	Cu	Pennant ss. (4/20)	—					(*Paterson Reversal* 2*m*)		
				C	Sediments U.K. (4/20)	M				Cu	Paterson Toscanite	—	
				C	Southdean basanite (4/22)	+							
				Cm	Tideswelldale rocks (4/20)	+	Cl–u	Sediments, Canada (1/108)	+	Cl	Lower Kuttung lavas	+	
				Cl	Derbyshire lavas (4/20)	M	Cl	Barnett Fm. (1/106, 107)	M				
	Cl	Tula horizon (5/51)	M	Cl	Carboniferous lst (5/47)	M	Cl	Codroy beds (1/109)	+				
	Cl	Tournai stage (5/52, 53)	M	Cl	Kinghorn lavas (4/20)	M							
	Cl	Siberian platform beds (5/54–60)	—	Cl	Wenlock lavea (4/20)	—							

to this occurs in a railway cutting at Paterson, N.S.W., between the Paterson Toscanite and the glacial sediments immediately overlying (Irving 1963b). This is referred to as the *Paterson Reversal.* The upper reversal in the Russian sections occurs either between the Lower and Upper Tartarian or within the latter, and may be represented in Australia by the *Illawarra Reversal* (section 3). It is proposed here to call the time span between the Paterson and Illawarra Reversals the *Kiaman Magnetic Interval*: the corresponding rock term proposed is *Kiaman Magnetic Division.* This name is put forward on historical grounds since it was in a sample from the Upper Marine latites collected near Kiama, N.S.W. that the French scientist, Mercanton, first reported the occurrence of positive directions for rocks of this period. "Une cube donne sans ambiguite: inclinaison boreale, 87°." Mercanton 1926, p. 189.) Although as more data is obtained there is likely to be much debate as to the exact stratigraphic position of the lower and upper reversal levels we suggest the term Kiaman to avoid confusion with geological terminology. It would be incorrect to use the word Permian, since the beds with uniform positive polarity are not confined to that system but extend down into the Carboniferous.

Eventually it may be possible to erect a time scale based on palaeomagnetic characteristics by the use of reversals and other observed effects (axis changes for example) useful as world-wide marker horizon. The possible correlation of the Illawarra Reversal with the reversal in the Tartarian of the U.S.S.R. is, for example, a point of potential importance for intercontinental correlation. A whole sequence of new names may be needed, and to avoid confusion it is desirable that these should be defined as above in terms of observations in the standard geological sequences in each continent otherwise much confusion will ensue.

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